STOCK PRICE TREND ANALYSIS

Problem Statement

The project analyses the relative decline/progress of the top 5 sectors within the NIFTY 50 group of companies. The pattern of their rise or fall over a period of 20 years guides retail buyers towards choosing the most stable company to invest their money in.

Key Objective

The key objective is to highlight the performance of the 5 key sectors (and the companies within them) with the highest contribution of market cap to the NIFTY 50 over a period of 20 years.

Who will use the dashboard

Investors and retail buyers can use this dashboard to assess the past performance of the top 50 stocks (by market cap). Further, traders can predict patterns and trends using the line graphs to foresee a company's prospective growth. This dashboard can also be used by financial analysts and the executive boards of companies to anticipate future crashes in the stock market and plan how to better equip the company to recover from these losses.

Data Description

Data Size: There are 5000 rows (on an average) and 15 columns in each of the 15 sheets of data. Within each of the 5 major sectors (Financial, IT, Energy, Consumer Goods and Metals), I have chosen the top 3 stocks (by market cap).

Categories:

1. Date: The date on which the data was recorded.
2. Symbol: The symbol of the company as used by the NSE.
3. Series: A code assigned by stock exchanges to dictate the specific trading and settlement rules that apply to them. (for example, EQ-equity trading)
4. Previous Close: The previous day's close price.
5. Open: The opening price of the stock.
6. High: The highest price reached by the stock (on that day).
7. Low: The lowest price reached by the stock (on that day).
8. Last: The last traded price in a day.
9. Close: The closing price of the stock.
10. VWAP: Volume Weighted Average Price shows the average price a security has traded at throughout the day, based on both price and volume.
11. Volume: The total number of shares traded (buying and selling) during a trading day.
12. Turnover: The number of times a business sells and replaces its stock of goods.
13. Trades: The count of how many separate times a transaction occurred.
14. Deliverable Volume: The number of shares bought and moved into permanent storage at the end of the day.
15. % Deliverable: The percentage of traded shares that are kept for long-term investment by traders (ratio of deliverable volume to total volume).

Data Preprocessing

1. Null Data
   There was null data in multiple columns. The blank cells were replaced by "N/A" to be then classified as a different category for the purpose of charts.

2. Filters
   I applied filters to isolate the null cells (blanks) in the columns and then label them as "N/A". Due to the large number of null cells, I chose to include them in the data instead of deleting them. I also applied value filters to sort stocks alphabetically. To highlight the top companies in each category, I used Conditional Formatting (top 10 rules).

3. Data Type
   Most of the columns contained general/text data type. I changed the format of the numerical data in the columns into Number format. I adjusted the decimal points accordingly, as per discrete and continuous values (mostly to two decimal places)

4. Formulas Used
   I used VLOOKUP to match the companies in the NIFTY 50 with their respected market caps as of December 2021. I imported the market caps into a different sheet from the NSE website as it was not previously included in the data. I also used INDEX + MATCH for this same purpose. I also used SUM to calculate total market cap of all the companies.

5. Additional Columns
   Data sheets for certain companies contained multiple symbols in the Symbol column. For example, Tata Steel initially went by TISCO before it as changed to TATASTL. Thus, to maintain uniformity, I added an extra column called "Previous Symbol" where I copied the entries of the "Symbol" column. After this, I replaced all the entries in the "Symbol" column with a common entry (for Tata Steel, I substituted TISCO for TATASTL).

6. Pivot Tables
   I used Power Query to connect three sheets at a time to make a collective Pivot Table to analyse three different stocks from the same sector. I did this for each of the five sector selected.

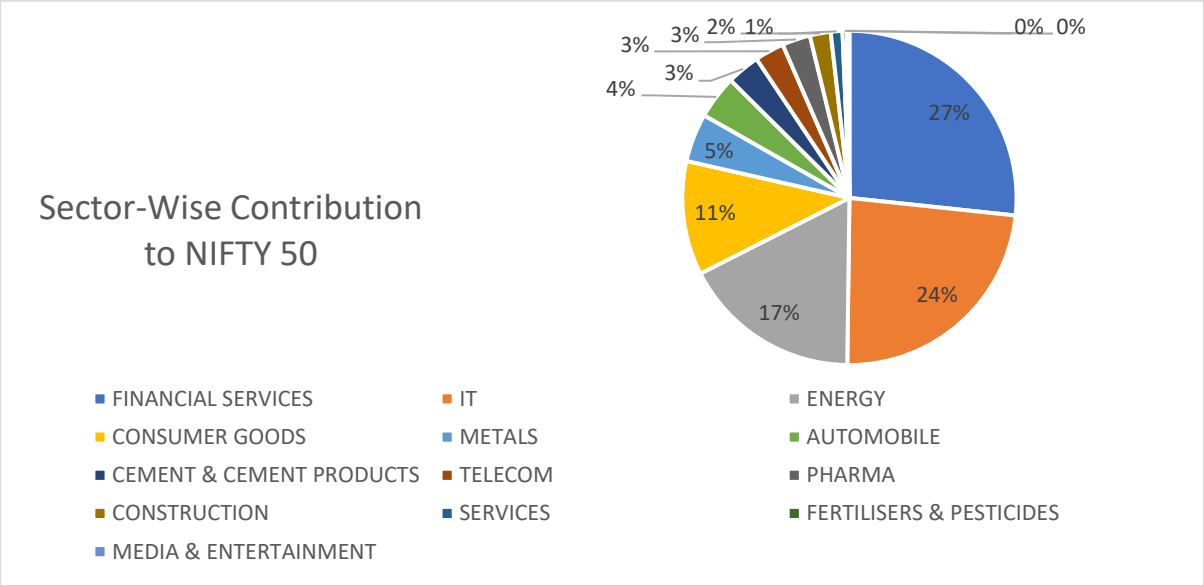
DashBoard Design

1. KPI Cards: Total number of companies analysed, total market cap (as of Dec 2021), largest sector (based on market cap), top company (based on market cap) and total market cap contribution of the leading companies in the top 5 sectors.
2. Pie Chart: Percentage contribution of each sector to the total market cap of the 50 companies analysed.
3. 2-D Bar Chart: Contribution of each company to the overall market cap of the 50 companies analysed.
4. 2-D Stacked Column Chart: Contribution of each company to the market cap of the sector it belongs to.

5. Line Chart: Measures closing stock prices of 5 leading companies (in the top sectors) daily for a period of 20 years.

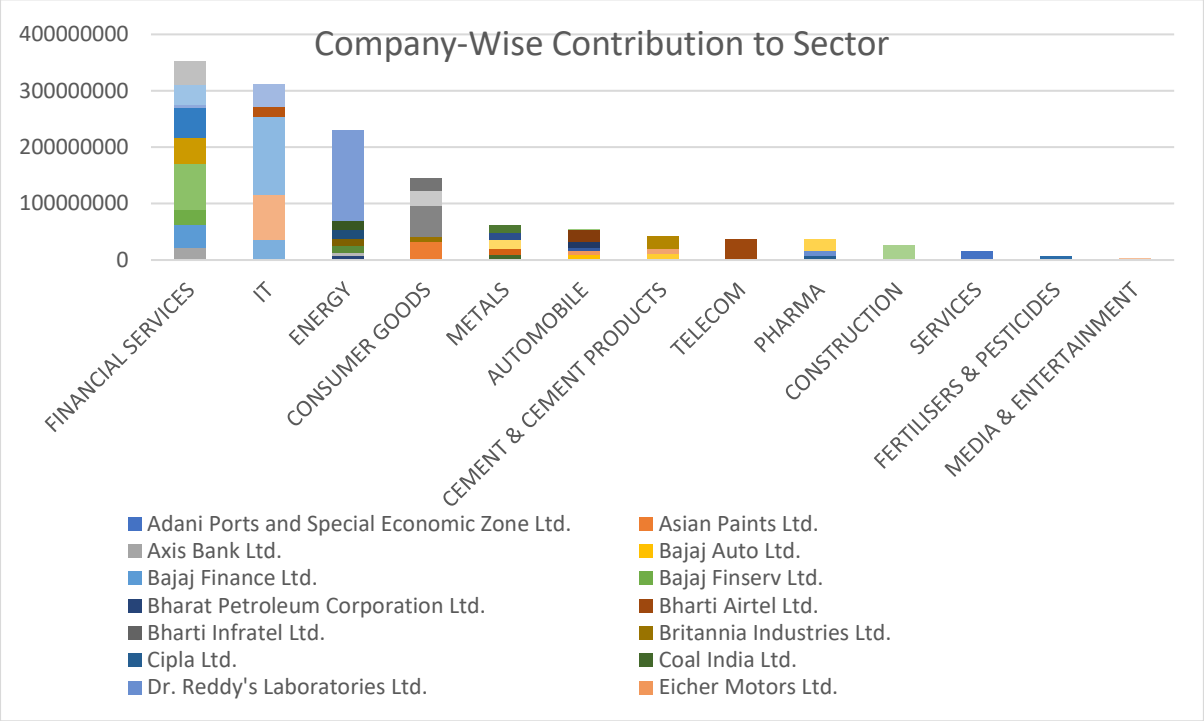
Key Insights

1. Sector-Wise Contribution to NIFTY 50



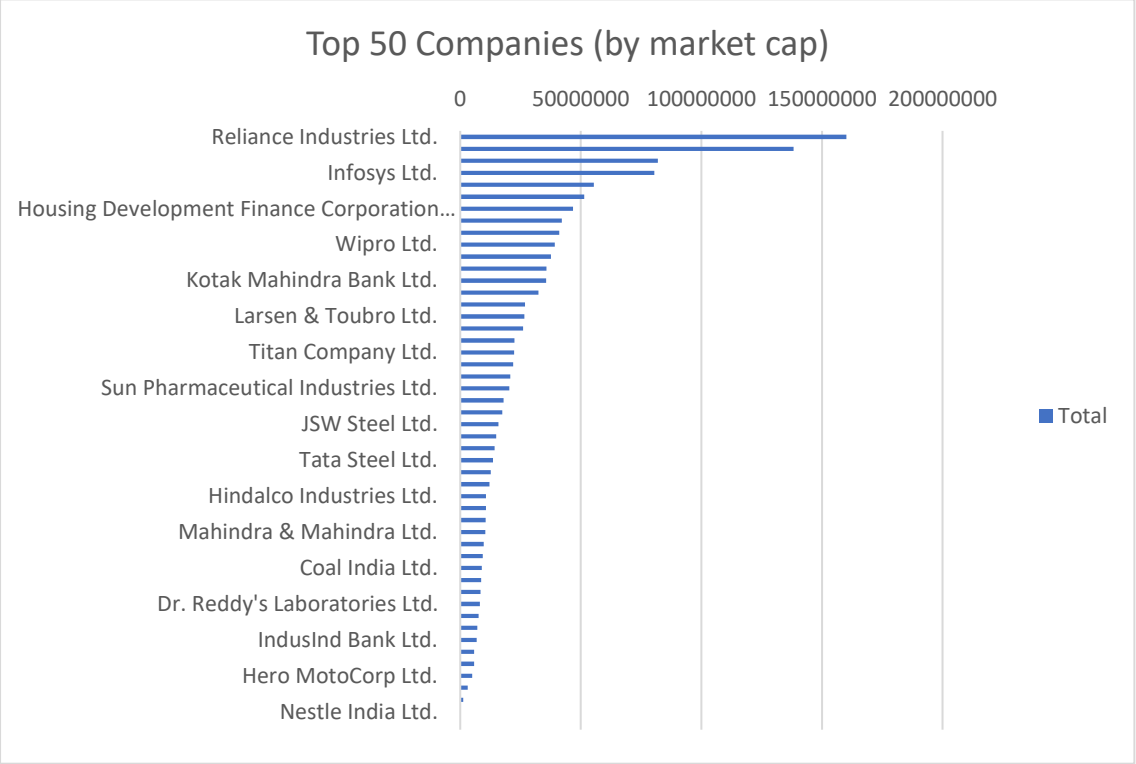
Financial Services and IT together account for approximately 50% of the total market-cap contribution represented in the analysis, making them the two dominant sectors. Energy follows at 17%, while Consumer Goods and Metals contribute 11% and 5% respectively.

2. Company-Wise Contribution to Sector



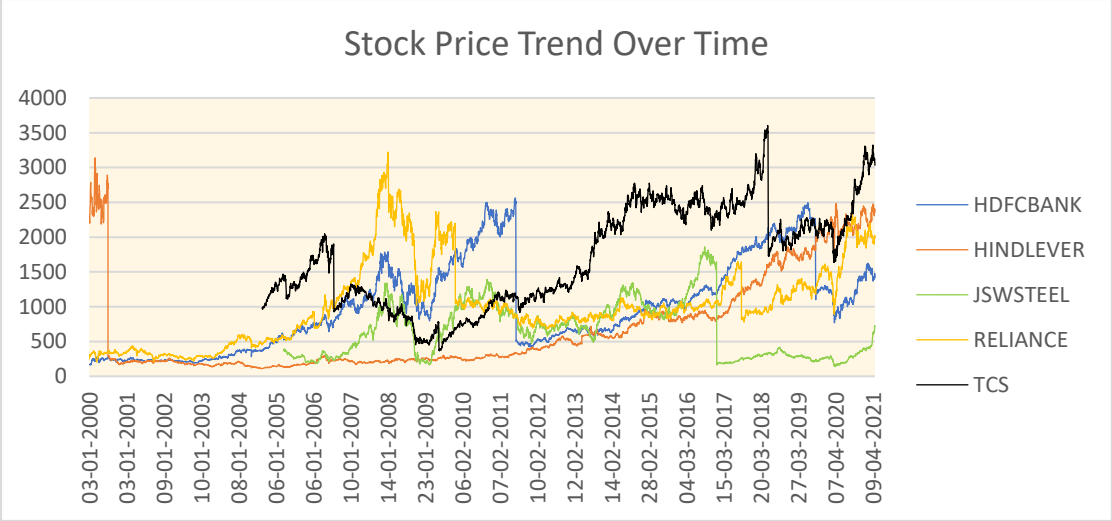
Market capitalization is unevenly distributed within several sectors, with a few companies accounting for a substantial share of their sector's total. Reliance dominates the Energy sector, while TCS and Infosys are major contributors within IT and HDFC Bank is the largest contributor within Financial Services. JSW Steel and Hindustan Unilever dominate the Metals and Consumer Goods sector respectively.

3. Top 50 Companies (by market cap)



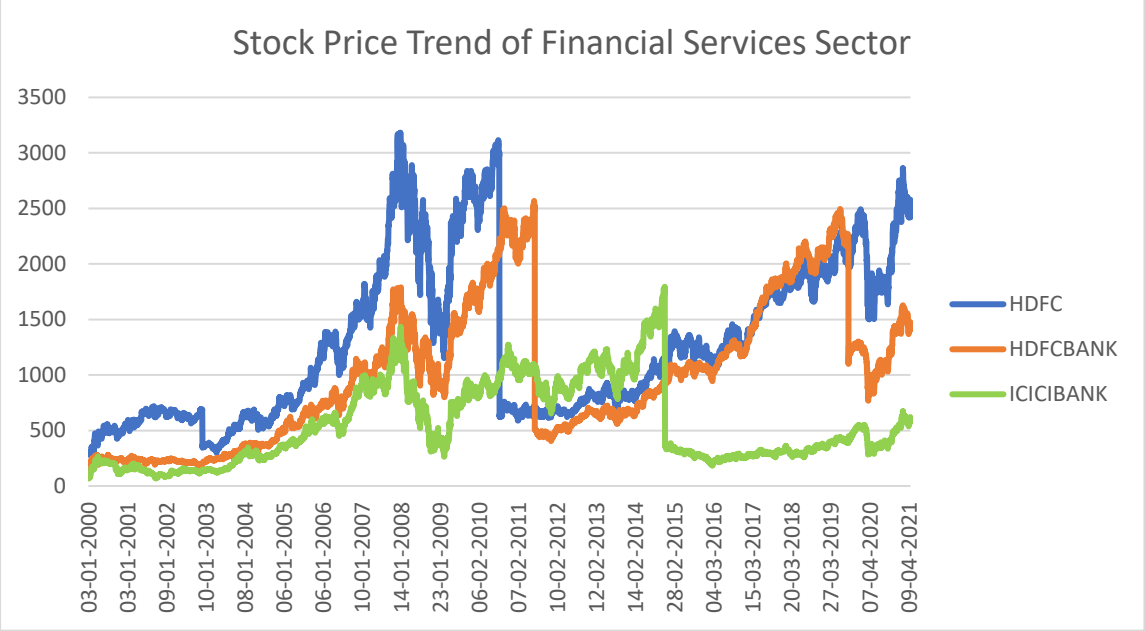
Reliance Industries and TCS have the largest market-cap contributions among the companies analysed, followed by HDFC Bank and Infosys. The gap between the top two companies and much of the remaining group indicates a relatively concentrated distribution of market capitalization.

4. Stock Price Trend Over Time (Sector-Wise)



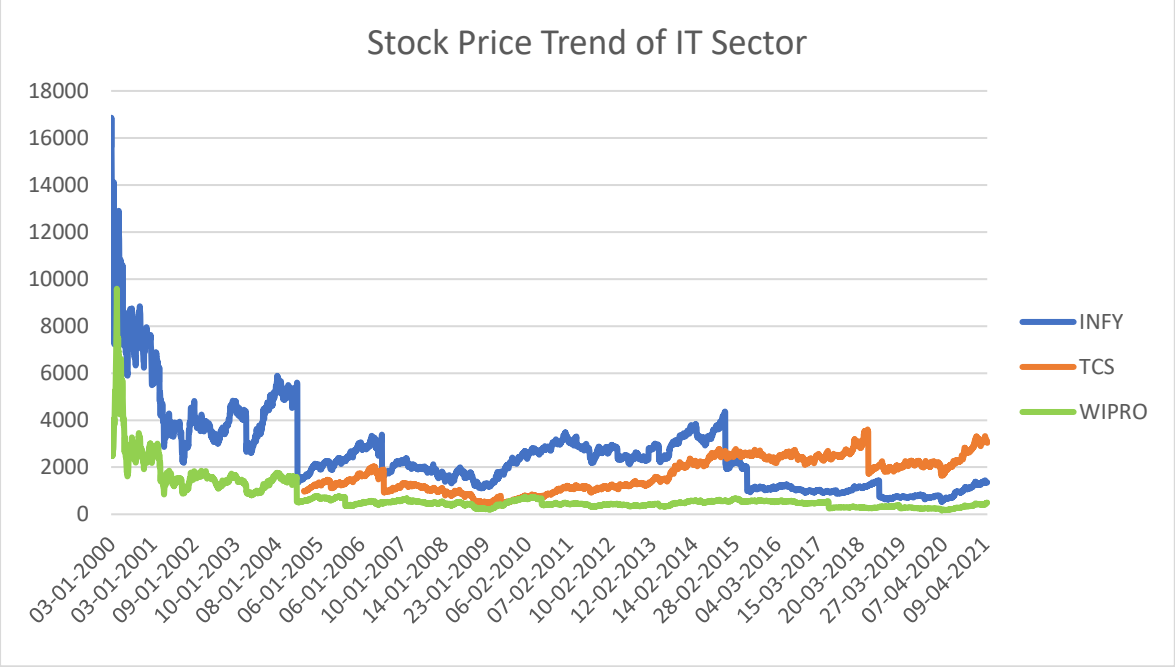
The five stocks display substantially different long-term price trajectories. HDFC Bank, Reliance and TCS show strong increases in their closing prices over the period, while JSW Steel experiences greater fluctuations and Hindustan Unilever's raw price remains comparatively stable toward the end of the period.

5. Stock Price Trend of Financial Services Sector



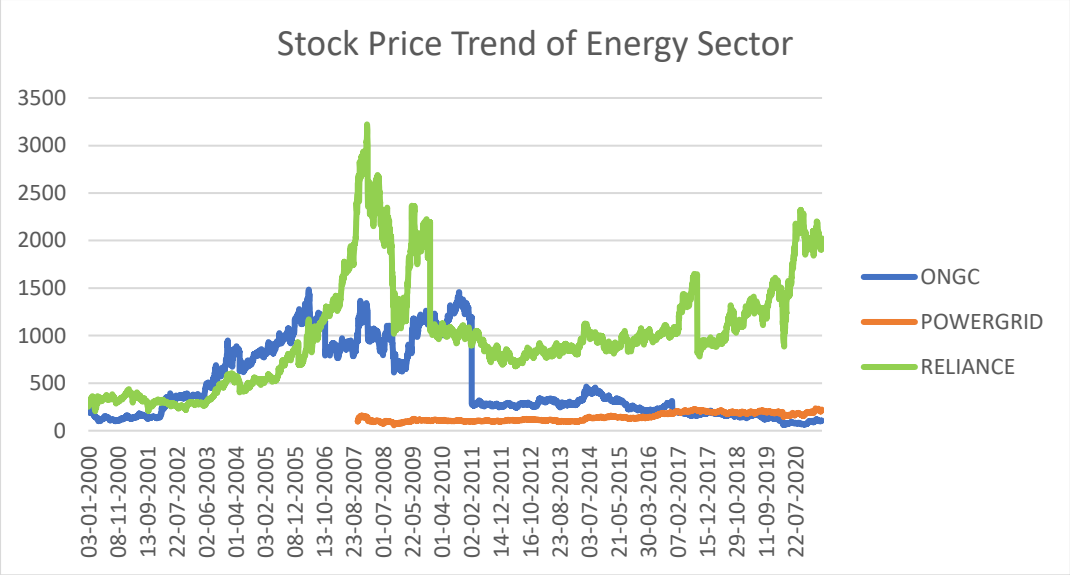
All three financial-services stocks show substantial long-term increases in closing prices, although each experiences significant fluctuations and periods of decline. The three stocks therefore exhibit a broadly upward long-term trajectory despite short- and medium-term volatility. HDFC recovers from a significant drop to emerge as the leading company.

6. Stock Price Tend of IT Sector



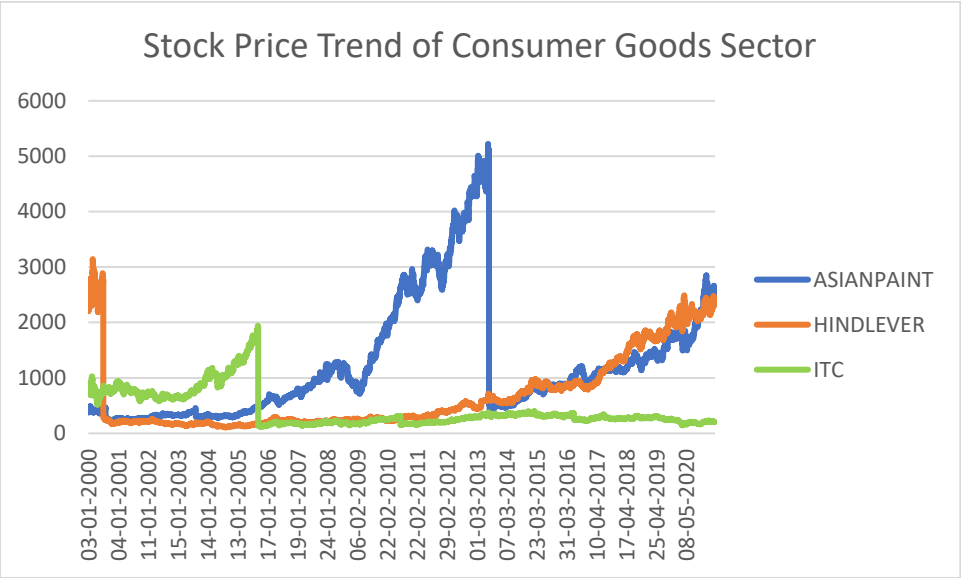
The IT stocks show markedly different price trajectories. TCS demonstrates sustained long-term growth, while Infosys and Wipro show much lower closing prices at the end of the period relative to their early observations, highlighting substantial variation within the sector.

7. Stock Price Trend of Energy Sector



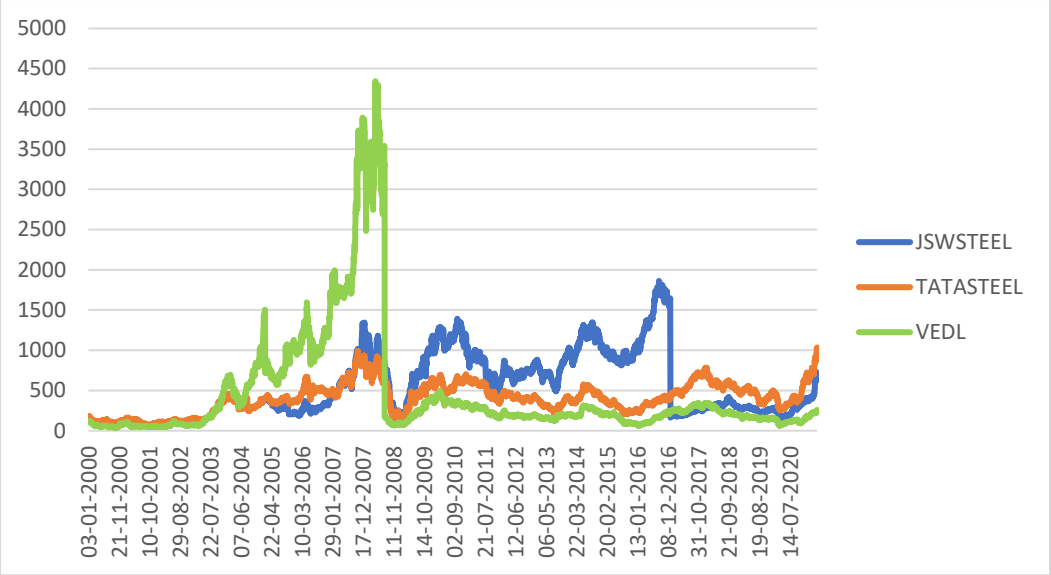
Reliance shows the strongest upward price trajectory among the three Energy stocks, while Power Grid records a moderate increase and ONGC ends below its starting price after considerable fluctuations.

8. Stock Price Trend of Consumer Goods Sector



Asian Paints shows a strong long-term increase in its closing price, while Hindustan Unilever remains relatively stable toward the end of the period and ITC finishes substantially below its initial recorded price.

9. Stock Price Trend of Metal Sector



The Metals stocks exhibit considerable price fluctuations over the 20-year period. Tata Steel shows a strong long-term increase, while JSW Steel and Vedanta experience pronounced peaks and declines before ending higher than their initial recorded prices.

Future Insights

Stock price alone does not provide a complete picture of company performance. Future research could incorporate P/E ratio, market capitalisation, dividend yield, earnings growth and trading volume to create a more comprehensive assessment of companies. One could also calculate daily return, intraday return, volatility and other parameters that would offer different measures of comparison.

The historical dataset could serve as a foundation for more advanced analysis using moving averages, trend analysis, correlation and eventually statistical or machine-learning models. These could investigate relationships in the data without relying solely on visual observations.

The 20-year analysis can help identify sectors and companies that have demonstrated relatively consistent long-term growth despite periods of market volatility. This suggests that future investors and analysts can examine long-term trends rather than relying only on short-term price movements.

Conclusion

The 20-year analysis of major NIFTY 50 sectors demonstrates that stock-market performance varies considerably across companies and industries over time. The data highlights the importance of examining long-term trends rather than focusing exclusively on short-term fluctuations. Differences in stock performance also show that sector and company leadership can change over time, making diversification and continuous analysis important. Overall, the project demonstrates how historical financial data can be transformed into meaningful insights and provides a foundation for deeper analysis of risk, growth and market behaviour.

Sources

https://www.kaggle.com/datasets/rohanrao/nifty50-stock-market-data?select=VEDL.csv

https://www.nseindia.com/static/regulations/listing-compliance/nse-market-capitalisation-all-companies